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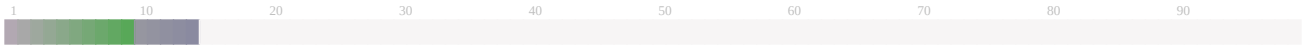
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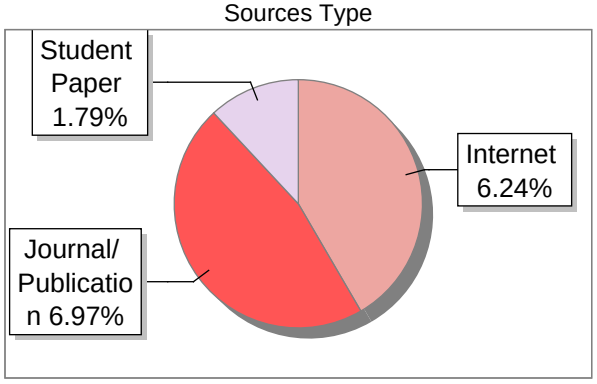
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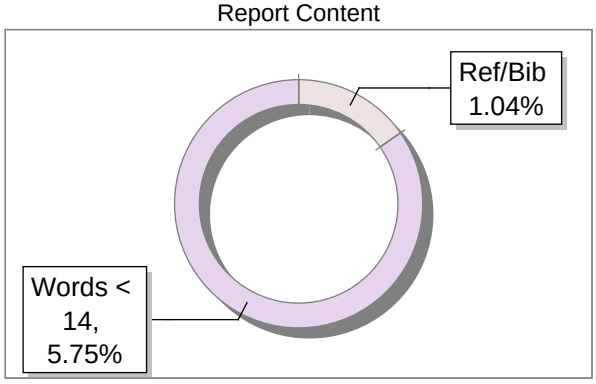
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Explainable AI-Based Clinical Decision Support for Preliminary Dermatological Disease Diagnosis

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***Abstract*— *Skin diseases are widespread and might be hard to diagnose correctly since many diseases look alike. Modern studies have proved that deep learning models can help with the classification of dermatological images, and explainable artificial intelligence could give a deeper look into the predictions [1], [3]–[5]. However, current solutions mainly pay attention to classification accuracy whereas clinical usefulness, interpretability, real-time access, and skin tone-related performance should also be considered [1], [2].*

This paper discusses Explainable AI-Based Clinical Decision Support System for Preliminary Dermatological Disease Diagnosis. The proposed system will be used dermatological images as inputs, applying deep learning techniques for diseases' classification. Preprocessing of the images and adequate deep learning architectures such as CNN-based or transfer learning will be used for extracting valuable features and making the prediction of the disease. Furthermore, the system will include Explainable AI methods such as Grad-CAM to show which regions play a key role in making the prediction and providing a confidence score to make it more interpretable. Web-based interface is proposed to provide access to the system.

The main aim of the proposed system is to support professional medical diagnostics and not to replace them. In addition to classification accuracy, the proposed system will consider interpretability and usability. Where suitable skin-tone information is available, model performance across different skin-tone groups will also be examined. The purpose of the research is to create a practical and transparent AI-driven system for preliminary dermatological analysis.

Keywords—
Skin Disease Classification, Deep Learning, Clinical Decision Support System, Explainable Artificial Intelligence, CNN, Grad-CAM, Dermatological Image Analysis, Transfer Learning, Skin-Tone Fairness, Preliminary Diagnosis.

I. INTRODUCTION

Dermatological diseases are widespread among the world's population and the identification of such conditions is important for further treatment. Traditional approach of dermatological diagnostics depends on the visual examination and the experience of professionals, however, sometimes there are no opportunities to receive assistance from dermatologists in rural or underserved areas. These issues are one of the reasons for using AI and

deep learning technologies for dermatological image analysis [1], [2].

Deep learning models have proven themselves as good classifiers for analyzing dermatological and dermoscopic images. Convolutional Neural Networks (CNNs) allow automatically learning visual features like texture, color, and shape, reducing dependency on manual feature extraction [9], [10]. The availability of open public databases like HAM10000 has additionally boosted research in this area. HAM10000 database consists of 10,015 dermoscopic images and covers multiple types of pigmented skin lesions [11].

However, modern approaches have gone beyond traditional CNN-based classification methods and investigated transfer learning, feature fusion, attention mechanism, EfficientNet, and transformers to improve dermatological images representation and classification [3], [4], [6], [7]. Also, multimodal approaches have examined the possibility of using image and clinical data to diagnose diseases [6]. Nevertheless, the improvement of classification accuracy does not make a model ready for use in clinical practice.

One of the biggest issues for deep learning-based medical systems is low interpretability of predictions. The model can predict the disease without showing the evidence of its choice. Explainable Artificial Intelligence (XAI) methods like Grad-CAM and LIME can provide the information about image regions or features, which play a key role in the model predictions [3]–[5]. This information makes possible to analyze the AI output and can enhance its usefulness.

Clinical usefulness of AI systems is another aspect worth considering. Previous studies have proved that AI-based decision support system can assist healthcare professionals in dermatological diagnostics and also investigated real-time systems for point-of-care applications [1], [2]. Meanwhile, it was found that AI performance differs depending on patients' skin tone, thus, high classification accuracy does not guarantee good performance for all users [1]. It is designed to provide disease predictions along with confidence levels and visual explanations through a web-based interface. Where suitable skin-tone data are available, the performance of the model across different skin-tone groups will also be evaluated. The system is not intended to replace professional diagnosis but to serve as an interpretable AI-based decision-support aid. Considering the above statements, this paper proposes Explainable AI-Based Clinical Decision Support System for Preliminary Dermatological Disease Diagnosis. The system will investigate deep learning-based classification of skin diseases using CNN and EfficientNet models and XAI to give visual interpretation of the prediction. It is designed to show the preliminary disease prediction with confidence information and explanation using web-based interface. The goal of the system is not to replace professionals' diagnosis but to provide an interpretable and accessible AI-based diagnosis

II. LITERATURE GAP

Some recent studies have investigated deep learning-based skin disease classification. CNN-based approaches have showed that deep learning allows learning valuable visual features from the skin lesion images, and publicly available databases like HAM10000 can be used for training and testing of the models [9]–[11]. More recent papers have looked into feature fusion, EfficientNet, attention mechanism, and transformer-based architectures to enhance the feature representation and classification [3], [4], [6], [7].

Even though these approaches provide strong classification accuracy, classification accuracy itself is not enough for a clinically useful system. The prediction of the model without any evidence is hard to analyze for users and healthcare professionals. The recent researches have tried to integrate XAI approaches like Grad-CAM and LIME to show the regions and features responsible for predictions [3]–[5]. However, the explainability is mostly considered as a separate component of the model, and is not always integrated into the whole prediction process.

Clinical application is another important issue to take into account. Point-of-care and AI-assisted dermatology systems have shown the ability of AI

to assist clinicians in the diagnostics [1], [2]. Similarly, clinical decision-support systems have

necessary, where possible, to evaluate performance in each of the different skin-

investigated the opportunity of using AI in primary-care dermatology settings [8]. At the same time, these studies have also proved that there are factors except classification accuracy to consider during practical deployment of AI.

Skin tone-related performance is another issue that should be considered. Modern clinical research has shown that AI-based decision support improves the diagnostics performance but still shows the differences in performance among different skin tone groups [1]. It means that high accuracy score does not guarantee the consistency of the performance for all users. Therefore, the analysis of the model behavior in different skin tone groups is necessary to build a more reliable dermatological AI system.

Moreover, multimodal approaches have shown the ability of combining dermatological images with clinical information, and lightweight architectures have been examined for practical usage [6], [7]. However, building of a single preliminary decision support system that will provide all these functions is still difficult.

Research Gap Identified

Considering the above review, the following gaps have been identified:

1. **Prediction-focused systems:** Current approaches mainly focus on optimizing disease classification performance [3], [4], [9].
2. **Limited integration of XAI:** Even though Grad-CAM and LIME are able to give explanations, the explainability is not consistently integrated into the prediction and interaction process [3]–[5].
3. **Clinical usability:** Current clinical approaches show the potential of using AI in dermatology, but preliminary assessment systems need to be developed [1], [2], [8].
4. **4. Skin-tone performance:** The level of classification accuracy alone cannot provide insight into how performance varies for different skin-tone groups. It is therefore tone groups [1].
5. **The integration of multiple components:** Classification, confidence information, visual explanation, and decision-support functionality are often investigated separately and not combined into one system.

Motivation of the Proposed Work

Considering these gaps, this paper aims at developing Explainable AI-Based Clinical Decision Support System for Preliminaries Dermatological Disease Diagnosis. The system will combine deep learning-based classification, confidence information, visual explanations, and a web-based interface. Alongside accuracy in classification, the current project will take into account both interpretability and usability. If skin tone information becomes available, performance will be assessed separately for different skin tones.

III. PROBLEM STATEMENT

Current AI-based dermatological systems have already proved to be very successful in diagnosing various skin diseases based on their images. Nevertheless, many of them concentrate mostly on high classification accuracy but do not provide enough information on why one certain prediction was made. This lack of explanation could be challenging for people to estimate the reliability of AI's output [3]–[5].

Moreover, according to researches related to the field of clinical decision support, AI can be used as a supportive tool in diagnosing dermatological diseases, yet there are some other aspects that should be taken into consideration in order to implement it in practice [1], [2], [8]. Among them are understandable explanations, confidence information, accessibility, and performance across different skin-tone groups [1].

Thus, the problem that needs to be solved in this work is the absence of the integrated and accessible system that combines dermatological disease classification with interpretable prediction and preliminary decision-support information. The proposed work would like to solve this problem by integration of deep learning-based image classification with Explainable AI and

The applied methodology includes preparation of the dataset, image preprocessing, deep learning-based classification, and model evaluation. The current implementation is aimed at binary classification of dermatological images into benign

classification with Explainable AI and development of the web-based Clinical Decision Support System.

The system would provide preliminary disease prediction, confidence information, and visual explanation of the prediction along with the statement that the output is not to be used instead of professional medical diagnosis. The model performance in different skin-tone groups would also be considered in this work.

IV. OBJECTIVES

The main objectives of the proposed work include:

1. The development of the deep learning-based system for the preliminary classification of dermatological diseases based on skin lesions' images.
2. The preparation and preprocessing of dermatological images from publicly available datasets for efficient training and evaluation of the model.
3. To investigate the use of EfficientNet-B0 for binary classification of dermatological skin lesion images.
4. The implementation of Explainable AI techniques (e.g., Grad-CAM, LIME) for obtaining understandable visual explanations of the prediction.
5. The provision of confidence information along with the predicted disease category.
6. The development of the web-based Clinical Decision Support System.
7. To measure the effectiveness of the models using accuracy, precision, recall, F1-score, and confusion matrix.
8. To assess model effectiveness for different skin tones, where suitable skin-tone data exists.
9. To make sure that the suggested system works as a support system and does not take the place of dermatological diagnostics.

V. PROPOSED METHODOLOGY

classification of dermatological images into benign and malignant classes based on a pretrained EfficientNet-B0 model. The application of explainable AI and web-based decision support systems will be the next stages of development of the system.

A. Dataset Collection and Preparation

The experiments involved dermatological images taken from the ISIC dataset. The prepared dataset was split into training, validation, and independent test sets. In total, 11,571 images were used, including 6,943 images for training, 2,314 images for validation, and 2,314 images for testing.

Each image had been annotated with a binary indicator of the class to which the skin lesion belongs (0 – Benign and 1 – Malignant). The training, validation, and test datasets have been separated to evaluate the performance of the model on unseen images.

B. Image Preprocessing and Augmentation

Before feeding images into the deep learning network, all images were converted to RGB format and resized to 224 × 224 pixels, as it matches the input size of EfficientNet-B0.

For the training dataset, data augmentation has been used to increase generalization capacity of the model and avoid overfitting. The types of augmentation used were:

- Random horizontal flipping
- Random vertical flipping
- Random rotation (up to 15 degrees)

After that, all images were converted to tensors and normalized by ImageNet mean and std values.

For the validation and test datasets, only resizing, tensor conversion, and normalization were used to make the evaluation unbiased (i.e., no random augmentation).

C. Deep Learning-Based Classification

For binary classification of skin lesions, EfficientNet-B0 has been used with ImageNet-

The trained model and decision-support part will be implemented in a web-based user interface. A user will be able to input the dermatological image and receive the corresponding preliminary assessment, confidence information, and XAI visualization.

The output of the system will be clearly communicated as a preliminary decision-

pretrained weights. The initial classification head of EfficientNet-B0 has been changed by a fully connected layer with 2 output units for benign and malignant classes.

In order to handle class imbalance in the training set, class-weighted cross-entropy loss has been used. Class weights are 0.615 (benign class) and 2.683 (malignant class).

The training process was performed by means of the AdamW optimizer with the learning rate equal to 0.0001 for five epochs using a batch size of 16 on a NVIDIA Tesla T4 GPU via Google Colab.

The model that achieved the highest F1-score on validation was selected as the best model and tested on an independent test set.

D. Explainable Artificial Intelligence

The selected classification model will be enhanced using Explainable AI techniques. For that purpose, Grad-CAM will be used for generation of heatmaps indicating the image areas responsible for particular prediction [5].

LIME can be also considered for generating local explanations for each prediction [3]. The explained model will be used to decide whether the algorithm is paying attention to the right lesions on images.

E. Confidence and Decision-Support Layer

The classification output will be processed for providing the disease prediction and its confidence information. The system will communicate the prediction as an initial AI-driven assessment rather than a diagnostic one.

The predicted class, confidence information, and XAI visualization will be processed in the decision-support layer because of existing clinical AI research that demonstrates the possibility of using AI as a supporting technology [1], [2], [8].

F. Web-Based System Integration

us to understand how the model is sure of the suggested class and not consider the prediction as the ultimate diagnosis.

E. Explainable AI

The prediction is processed in the XAI module, where Grad-CAM provides the heatmap showing which areas of the image the model used to make the prediction. In case it is needed, the local

communicated as a preliminary decision supporting data and not a substitution for professional medical diagnostics.

G. Performance and Fairness Evaluation

The performance of developed models will be estimated by accuracy, precision, recall, F1-score, and confusion matrix analysis.

When possible based on the chosen evaluation dataset, the models will be tested on performance fairness for different skin tones. It is especially important because of existing clinical research that shows that general improvement in AI-assisted diagnostics doesn't guarantee equal performance across different skin tones [1].

VI. SYSTEM WORKFLOW

The proposed system has a sequential workflow from inputting the image till obtaining an interpretable preliminary assessment. The major stages of the workflow are presented below.

A. Image Acquisition

The user uploads an image of dermatological conditions or skin lesions via the web interface.

B. Preprocessing of Image

The uploaded image is resized and normalized to comply with the specifications for the input into the chosen deep learning model. Suitable preprocessing steps are taken.

C. Feature Extraction & Disease Classification

The preprocessed image is then fed to the trained model, which extracts features and classifies the disease category with maximum probability.

D. Confidence Estimation

Along with the classification result, there is information about the level of confidence. It allows

↓

Web-Based Output

The above-mentioned flow is the key value of the proposed system because it combines the prediction and all the supporting information.

VII. TOOLS AND TECHNOLOGIES

The proposed system requires some software tools

explanation may be provided via LIME.

F. Clinical Decision Support

All three pieces of information (the prediction itself, confidence information, and explanation) are combined in order to produce the preliminary decision-support output. It is expected that the system will help the user to get preliminary conclusions but not to replace a dermatologist.

G. Final Output

In the web interface, we have the following information:

- The predicted disease;
- The confidence information;
- The heatmap from the XAI module;
- The preliminary decision-support information;
- The medical disclaimer stating the need for the professional consultation.

Overall Flow

User Image

↓

Preprocessing

↓

CNN / EfficientNet Model

↓

Disease Prediction + Confidence

↓

Grad-CAM / LIME

↓

Clinical Decision Support Layer

and libraries for image preprocessing, deep learning model development, explainability, visualization, and web-based deployment. The main tools which are supposed to be used for implementation are presented below.

Category	Tool / Technology	Purpose
Programming Language	Python	Model development, data processing, and evaluation
Deep Learning Framework	PyToch	Model implementation and training
Deep Learning Model	EfficientNet-B0	Binary skin lesion classification
Image Processing	Pillow, Torchvision	Image loading, resizing, augmentation, and normalization
Data Processing	Pandas, NumPy	Dataset and numerical data handling
Evaluation	Scikit-learn	Accuracy, recall, F1-score, and confusion matrix
Environment	Google Colab, NVIDIA Tesla T4	Model training and GPU acceleration

VIII. EXPERIMENTAL SETUP

The experimental setup was developed in order to evaluate the results of implementation of the skin lesion classification model using deep learning approaches. Experiments were run in PyTorch environment using GPU in Google Colab. At the moment, the current experiments focus on binary classification of skin lesion images in terms of benign or malignant using EfficientNet-B0.

A. Dataset Preparation

The prepared ISIC dermatological images dataset was used in experiments, which comprised a total number of 11,571 images divided into training, validation, and independent test datasets.

Dataset distribution:

cross-entropy loss was used to handle this issue. The computed class weights are 0.615 for benign and 2.683 for malignant.

Training was done using the AdamW optimizer with the learning rate equal to 0.0001. The batch size was 16, and training lasted for 5 epochs. The model checkpoint with the highest validation F1-score was saved as the best-performing model and subsequently used for testing.

D. Evaluation Metrics

- Training set: 6,943 images
- Validation set: 2,314 images
- Test set: 2,314 images

Every image has a binary label, where 0 corresponds to Benign and 1 stands for Malignant. The training dataset was used to tune model parameters, while the validation set was used to control model performance and select the optimal model. The independent test dataset was left for the final evaluation of the trained model.

B. Image Preprocessing

All the input images were converted to RGB format and resized to 224 × 224 pixels as per the input requirements of EfficientNet-B0.

Data augmentation was done on training images, where random horizontal flipping, random vertical flipping, and random rotation up to 15 degrees were applied. Then the images were converted to tensors and normalized using ImageNet mean and standard deviation.

Validation and test data did not undergo any random augmentations, but they were just resized, converted to tensors, and normalized.

C. Model Configuration and Training

The pretrained EfficientNet-B0 model with ImageNet weights was used for binary skin lesion classification. The classifier of the model was substituted by the fully connected layer with 2 output neurons, which correspond to the benign and malignant classes.

Since the dataset is imbalanced, there are less malignant images than benign ones, class-weighted

Accuracy, precision, recall, F1-score, and confusion matrix were used to evaluate the performance of the model. The metrics were chosen in order to give a holistic evaluation of the model's classification performance and ability to detect malignant lesions.

Specific attention was paid to the evaluation of malignant-class recall and F1-score since the detection of malignant lesions is crucial in a clinical screening setting.

E. Experimental Environment

The model was implemented using Python and PyTorch in Google Colab environment. The computations were accelerated by means of NVIDIA Tesla T4 GPU.

Main libraries used in the experiments were:

- PyTorch to implement and train the model
- Torchvision to use EfficientNet-B0 and transform images
- Pandas to work with datasets and CSV files
- Pillow to load and process images
- Scikit-learn to calculate metrics

F. Skin-Tone Performance Evaluation

The analysis of skin-tone performance was not done in the current study due to lack of data on skin tone in the data set. Therefore, any difference in performance between skin-tones could not be established. This can serve as an area for further research in the suggested framework.

IX. RESULTS AND DISCUSSION

A. Classification Performance

The EfficientNet-B0 model was trained and later evaluated using a test dataset of 2,314 images. Test results revealed that test accuracy was 86.47%, malignant class recall was 84.92%, and malignant class F1-score was 70.05%.

This demonstrates the ability of the EfficientNet-B0 model to classify a significant number of benign and malignant skin lesion images. The malignant recall of 84.92% indicates that the model correctly identified a substantial proportion of malignant cases.

TABLE I

Although the model identified a substantial proportion of malignant cases, the 65 false-negative cases highlight the need for further improvement before clinical use.

C. Explainable AI Analysis

Analysis using the Explainable AI technique using Grad-CAM and LIME was not performed for the present experimental results. Therefore, no visual or numerical insights about the regions that were used by the model during the predictions can be reported at this point.

The integration of Grad-CAM and LIME might be used in the future for visualizing model

CLASSIFICATION PERFORMANCE OF EFFICIENTNET-B0

Model	Accuracy	Recall	F1-Score
EfficientNet-B0	86.47%	84.92%	70.05%

In the classification report, it can be seen that the benign class has 0.96 precision, 0.87 recall, and 0.91 F1-score, whereas malignant class has 0.60 precision, 0.85 recall, and 0.70 F1-score.

The difference in precision between benign and malignant classes demonstrates the fact that despite the ability of the model to detect many malignant cases, some benign cases were misclassified as malignant. It is illustrated in the confusion matrix.

B. Confusion Matrix Analysis

The confusion matrix for independent test data is presented below:

Confusion Matrix:

[[1635 248]

[65 366]]

The model correctly identified 1,635 benign images and 366 malignant images. It incorrectly classified 248 benign images as malignant and 65 malignant images as benign. Thus, 366 of the 431 malignant cases were correctly identified, corresponding to a malignant recall of 84.92%.

predictions and checking whether the model is paying attention to clinically meaningful regions in skin lesions.

D. Analysis of Skin Tone Performance

Performance analysis for different skin tone groups was not performed in the current experiment as skin tone data were not provided for the considered dataset.

Conclusions about the performance in different skin tone groups therefore cannot be made based on the results obtained in the current research. This task might be undertaken in future work based on the proposed system.

E. Decision-Support Information

In the current experiment, only the classification part of the proposed system was examined. The trained EfficientNet-B0 model makes binary predictions – either Benign or Malignant.

The full decision-support interface with information about confidence, XAI visualization, and decision-support information can be integrated in the system in the next stage of development.

F. Discussion

The EfficientNet-B0 model achieved 86.47% accuracy on the independent test set, with a malignant-class recall of 84.92% and F1-score of 70.05%. The confusion matrix shows that 366 of 431 malignant cases were correctly identified, while 65 malignant cases were incorrectly classified as benign. Although the model demonstrated good malignant-class recall, the false-negative cases indicate that further improvement is required. Since Grad-CAM, LIME, skin-tone fairness analysis, and the web-based decision-support interface were not evaluated in the current experiment, conclusions regarding explainability, fairness, and system usability cannot be drawn from the present results. These components remain directions for future development.

X. APPLICATIONS

The proposed system can be used in the following

The system can provide understandable information about preliminary assessment made by AI and explicitly direct patients towards professional medical consultation.

XI. CONCLUSION

From the above experiments, it can be seen that the proposed model for efficient binary classification of benign and malignant skin lesions is effective. In the independent test dataset consisting of 2,314 images, the model achieved an accuracy of 86.47%, recall of 84.92% for the malignant class and F1-score of 70.05% for the malignant class. The model was able to identify

areas:

1. Preliminary Skin Disease Screening:

The system can process the uploaded image of the skin lesion and make a preliminary prediction helping to identify the cases requiring professional attention.

2. Clinical Decision Support:

The predicted disease, confidence information, and visual explanation can serve as additional information assisting healthcare professionals in preliminary assessment.

3. Remote Healthcare Support:

A web-based system can perform preliminary AI-assisted dermatological assessment in those regions where there is no access to dermatologists.

4. Medical Education and Training:

Grad-CAM and LIME visualizations can be used as teaching materials explaining what regions of the image influence the prediction of AI model.

5. Evaluation and Research of AI Models:

The framework can be used in order to compare different deep learning architectures and investigate the role of Explainable AI in dermatological image classification.

6. Preliminary Patient Awareness:

malignant class. The model was able to identify 366 out of 431 malignant cases whereas 65 malignant cases were wrongly classified as benign cases. This shows room for improvement in the detection of malignant cases. This research provides a baseline for the proposed skin lesion classification system. As the project progresses, the skin-lesion classification system will be extended to improve malignant case detection capability, include explainability through Grad-CAM and LIME, evaluate skin tone fairness and finally include a web interface for clinical decision support.

XII. FUTURE WORK

The proposed system could be developed in the following directions:

1. Larger and More Diverse Datasets:

In future work, the system could be expanded to include larger and more diverse dermatological datasets to increase the generalization power of the model.

2. Multimodal Analysis:

It would be possible to extend the system through the combination of skin images with symptoms and clinical data to provide more thorough preliminary diagnosis [6].

3. Improved Fairness Evaluation:

In future studies, it will be possible to test the performance of the model in terms of different skin-tone groups [1].

4. Advanced and Lightweight Models:

In future studies, more advanced and lightweight models could be explored to decrease the requirements for computation power and increase the speed of inference [7].

5. Explainability:

In future studies, different approaches to explainability could be compared, and the extent to which their explanations refer to clinically relevant lesion regions could be studied.

6. Clinical Validation:

The system could be validated on independent clinical datasets in collaboration with

[4] Enhancing Skin Disease Classification Leveraging Transformer-based Deep Learning Architectures and Explainable AI

[5] Explainable CNN Framework with Grad-CAM Analysis for Real-Time Skin Disease Detection

[6] Attention-guided Multimodal Skin Disease Classification Using Discounted Dempster-Shafer Fusion and EfficientNet

[7] FUSCANet: Enhancing Skin Disease Classification Through Feature Fusion and Spatial-Channel Attention Mechanisms

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dermatologists.

7. Mobile and Cloud Deployment:

The web-based system could be transformed into mobile and cloud-based to improve its accessibility and provide real-time dermatological assessment.

8. Additional Clinical Guidance:

In the future, it would be possible to provide more structured preliminary guidance, for example, recommending professional consultation based on the confidence level of the model's prediction and its severity.

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